MirrorSentinel: Suppressing Mirror-Induced Ghost Structures in Elevator LiDAR Maps with Visual Depth

Yugong Wang[1], Shengzhe Gan[1], Yu Zhan[1], Chuyao Fu[2], Yitao Zeng[1], and Hong Zhang[1,*], *Fellow, IEEE*

***Abstract*— Highly reflective surfaces can produce coherent ghost walls in indoor LiDAR maps even when the estimated trajectory is accurate. We present MirrorSentinel, an offline map cleaner that uses registered LiDAR scans to calibrate monocular depth and then reprojects the completed map into historical camera views. Voxels are removed only when multiple views place a visible surface in front of them and provide little support for retaining them. The trajectory and coordinates of all retained points remain unchanged. On eight recorded sequences from seven measured elevators, MirrorSentinel reduces the exterior ghost footprint by 42.1% and raises wall F-score from 0.5448 to 0.5757, while boundary coverage changes from 0.8356 to 0.8353. The footprint reduction remains 31.5% after both maps are normalized to exactly 50,000 points. Under identical per-sequence rejection budgets, MirrorSentinel also achieves the highest F-score against random, statistical, radius, and single-view residual controls. These results show that LiDAR-calibrated monocular depth can remove persistent mirror artifacts without changing the trajectory or moving retained geometry. Our code and dataset are available at** `github.com/KvnWong216/MirrorSentinel`**.**

I. Introduction

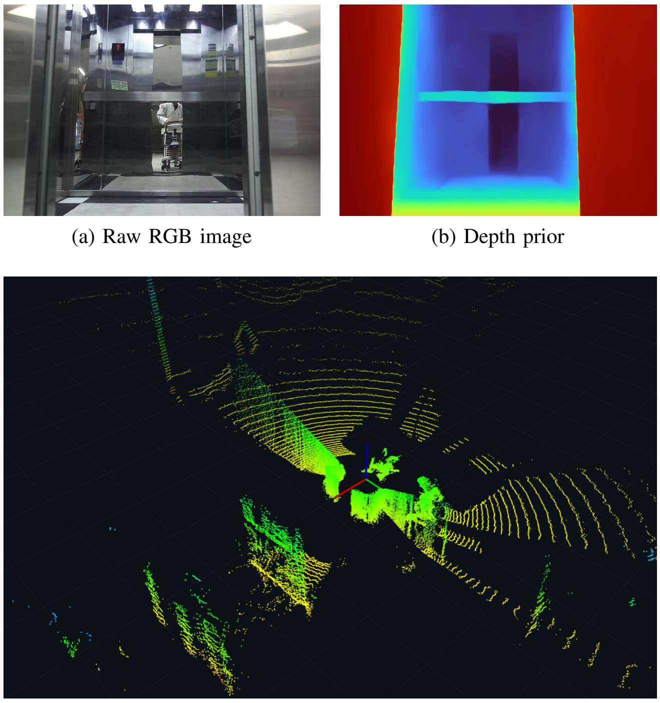
Large-scale indoor SLAM commonly combines a LiDAR, an inertial measurement unit (IMU), and often a camera to estimate motion and build a geometric map. LiDAR–inertial odometry (LIO) and LiDAR–inertial–visual odometry (LIVO) work well in many buildings, but highly reflective surfaces violate their usual direct-return assumption. A mirror-lined elevator is a severe example. A laser pulse can reflect inside the cabin and return along an indirect path, while the LiDAR places the measured range on the original beam. The resulting point appears behind the mirror. Across successive scans, such points register into dense, coherent ghost walls rather than isolated noise, contaminating maps used for localization, inspection, or navigation.

Improving the trajectory alone does not remove this structure. LIO and LIVO systems treat consistently registered endpoints as geometric support, so a coherent mirror ghost can satisfy the optimizer even when the trajectory is accurate [1]–[4]. Reflection-aware methods add reflector planes, intensity, multiple returns, angular response, or full waveforms [4]–[7]. These cues can expose the reflected path, but they may be unavailable on a robot carrying an RGB camera and a single-return LiDAR.

[1]Yugong Wang, Shengzhe Gan, Yu Zhan, Yitao Zeng, and Hong Zhang are with Shenzhen Key Laboratory of Robotics and Computer Vision, Southern University of Science and Technology.

[2]Chuyao Fu is with Robotic Perception and Intelligence Laboratory, Southern University of Science and Technology.

[*]Corresponding author (`hzhang@sustech.edu.cn`).



(a) Raw RGB image

(b) Depth prior

(c) LiDAR point cloud

Fig. 1. Mirror corruption in an elevator. Multipath LiDAR returns form coherent points behind the physical cabin. Monocular depth provides a complementary clue by indicating when the visible cabin surface lies in front of those mapped points.

Monocular depth offers a complementary cue by estimating the nearest visible surface along each camera ray. Its scale can be inaccurate and mirrors can confuse its prediction [8], so it should not replace LiDAR geometry. It can instead test an existing map point. If LiDAR-calibrated depth repeatedly places the visible cabin wall clearly in front of that point, the point lies behind a surface that should have blocked it. Observations from different robot positions help distinguish persistent evidence from a single depth error.

Based on this idea, we propose MirrorSentinel, an offline system for cleaning a completed LiDAR map. During replay, registered scans recover the scale of each usable depth frame. The completed map is then projected into images from different robot positions. A voxel accumulates evidence when it lies behind the visible surface, while depth agreement supports retaining it. Only voxels with sufficient removal evidence and weak support are deleted, without changing the trajectory or generating new 3D points.

MirrorSentinel uses a LIVO system for registered scans, poses, and the raw map, and a pretrained monocular model for dense depth. Our contributions are the robust per-frame LiDAR scale calibration that connects these sources, the multi-view voxel decision that balances removal and support evidence, and a same-map evaluation with density-normalized and per-sequence matched-budget controls. Across eight recorded sequences from seven measured elevators, the results show substantial ghost removal while preserving the measured cabin boundary.

II. Related Work

A. Why Conventional Mapping Retains Mirror Ghosts

LiDAR–inertial mapping spans smoothing-based systems such as LIO-SAM and direct, filter-based systems such as FAST-LIO2 and Point-LIO [9]–[11]. LVI-SAM, R3LIVE, and FAST-LIVO2 further couple images for state estimation, color, or mapping [12]–[14]. These systems still register measured ranges along their assumed direct paths. Mirror multipath violates that assumption. Indirect ranges can form coherent planes and become local-map support rather than sparse outliers [2], [15], [16]. Visual updates do not by themselves test the optical path of every endpoint. This motivates a post-mapping check that compares the completed map across the frozen trajectory.

B. Monocular Depth as a Geometric Cue

Metric single-image models and video predictors provide increasingly general dense depth [17]–[19], while recent systems target stronger multi-view consistency or efficient zero-shot inference [20]–[22]. Their scale and reflective-region accuracy remain uncertain [8], so MirrorSentinel does not replace LiDAR geometry. It calibrates each usable depth frame with registered LiDAR and uses only the ordering between the visible surface and an existing endpoint. Reflection masks provide appearance information [8], [23], but cannot determine whether a 3D return belongs to a reflector, transmitted object, or virtual image.

C. Reflection-Aware Mapping

Earlier methods classify reflective or transparent surfaces from return intensity and geometry, or explicitly trace virtual points behind glass [2], [15], [16], [24]. Recent reflector-plane approaches use incidence angle, intensity, or multiple returns [3], [4], [6]. Reflectance Field Map instead models view-dependent range behavior in 2D occupancy [5], whereas Ghost-FWL uses full waveforms [7]. Static-scan methods combine multi-echo plane geometry, ray tracing, or visual glass masks [25], [26]. These methods motivate our plane and mask controls but require sensing modes, reflector models, or outputs unavailable in our mobile single-return maps. MirrorSentinel instead uses the registered scans, poses, and RGB camera already present in a LIVO stack, without fitting a mirror plane or generating geometry.

III. Method

MirrorSentinel is an offline map-cleaning method that checks completed LiDAR maps against depth images recorded during mapping. FAST-LIVO2 supplies registered scans, poses, and the raw map. Paired scans first calibrate visual depth. The completed map is then projected into historical views, where agreement and contradiction votes determine removal. Retained points and the estimated trajectory remain unchanged.

A. Physical Model and Observable Cue

We begin with the mirror path described in Sec. I. Let o be the LiDAR origin, d the emitted unit direction, q the point where the beam meets the mirror, and x the physical scene point reached after reflection. The path-equivalent range and the endpoint reported along the original beam are

$$\ell_{\mathrm{mp}} = \|o-q\| + \|q-x\|, \qquad p_{\mathrm{virt}} = o + \ell_{\mathrm{mp}} d. \quad (1)$$

Thus $p_{\mathrm{virt}} \neq x$ and appears behind the mirror. Rather than estimate q or reconstruct x, we use the observable consequence of (1). A mapped endpoint is inconsistent with a direct return when the camera repeatedly sees a nearer surface in front of it. Multiple views are required because occlusion or transmission can create the same ordering once.

World, LiDAR, and camera frames are W, L, and C. T_{AB} maps frame B to A, $T_{LW} = T_{WL}^{-1}$, and T_{CL} is the LiDAR–camera extrinsic. FAST-LIVO2 provides $T_{WL}(t)$ and $M_{\mathrm{raw}} = \{p_m^W\}_{m=1}^N$. Scan $S_k = \{x_{k,i}^L\}$ is recorded at t_k^L. Camera-axis depth $D_j : \Omega \to \mathbb{R}_+$ is recorded at t_j^D. Valid inverse depth ξ_j is converted by $D_j = 1/\xi_j$.

The cleaning stage holds the SLAM result fixed:

$$T_{WL}^{\mathrm{clean}}(t) \equiv T_{WL}^{\mathrm{raw}}(t), \qquad M_{\mathrm{clean}} \subseteq M_{\mathrm{raw}}. \quad (2)$$

Here M_{clean} is the output map, so cleaning only removes endpoints. Frames and scans are paired one-to-one by minimum timestamp difference under synchronization tolerance $|t_j^D - t_k^L| \leq \Delta t_{\max}$, with poses interpolated at t_j^D.

B. Calibrating Visual Depth with LiDAR

The ordering test requires LiDAR and visual depth in one scale. Because even metric predictors can drift by scene, each frame is calibrated from its paired scan before it can vote.

For a scan point $x_{k,i}^L \in S_k$, its registered world position is $p_{ki}^W = T_{WL}(t_k^L) x_{k,i}^L$. We transform this fixed point into camera frame C at depth time t_j^D and project it into the image:

$$\begin{aligned} p_{kji}^C &= T_{CL} T_{LW}(t_j^D) p_{ki}^W, \\ u_{kji} &= \pi(K p_{kji}^C), \qquad z_{kji}^L = [p_{kji}^C]_z. \end{aligned} \quad (3)$$

Here K is the intrinsic matrix, π is perspective division, and (u_{kji}, z_{kji}^L) are the projected pixel and LiDAR camera-axis depth. We keep the nearest positive return per pixel and exclude borders. A correspondence near a mixed-depth boundary is also rejected when its local LiDAR-depth percentile range $q_{90} - q_{10}$ exceeds τ_e.

Let $\mathcal{C}_j$ contain the survivors, $n_j = |\mathcal{C}_j|$, and omit fixed k, j subscripts below. Each ratio $z_i^L / D_j(u_i)$ proposes a scale,

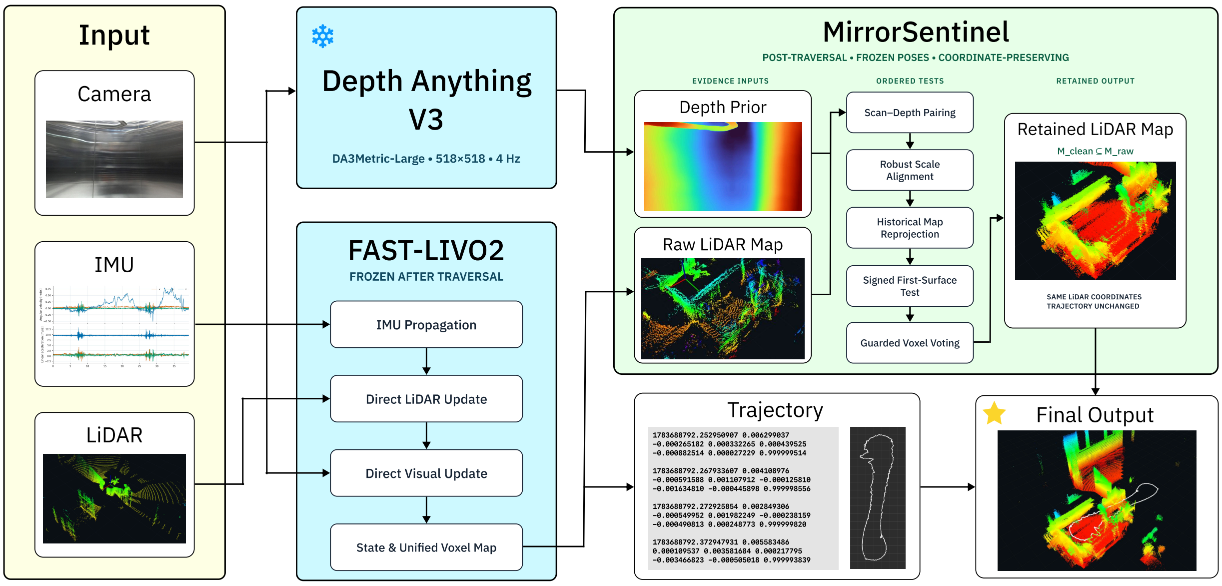


Fig. 2. Overview. FAST-LIVO2 supplies registered scans, M_{raw}, and frozen poses. Calibrated DA3 depth tests reprojected voxels across historical views. Voxels with repeated contradictions and weak support are removed without changing the trajectory or retained coordinates.

forming $\mathcal{A}_j = \{z_i^L / D_j(u_i) : i \in \mathcal{C}_j\}$. The clipped running log-scale mean $\bar{s}$ is added when available. We select the candidate with the largest correspondence consensus:

$$s_j^{(0)} = \arg\max_{s \in \mathcal{A}_j} \frac{1}{n_j} \sum_{i \in \mathcal{C}_j} \mathbf{1}\big(|z_i^L - sD_j(u_i)| \le b_i\big) - \lambda |\log(s/\bar{s})|, \quad b_i = \max(\gamma_a, \gamma_r z_i^L). \tag{4}$$

Here b_i combines absolute tolerance γ_a and relative tolerance $\gamma_r z_i^L$. The λ term discourages abrupt change from $\bar{s}$. Before a prior exists, both this term and the log-scale gate are omitted.

For residual $e_i = z_i^L - s_j^{(0)} D_j(u_i)$, a median absolute deviation (MAD) test removes remaining contamination:

$$|e_i - \text{med}(e)| \le \kappa\, 1.4826\, \text{MAD}(e). \tag{5}$$

where κ controls rejection. If $\mathcal{I}_j$ contains correspondences passing both tests, the calibrated scale is their median ratio:

$$s_j = \text{median}_{i \in \mathcal{I}_j} \frac{z_i^L}{D_j(u_i)}. \tag{6}$$

We repeat the intersection once around s_j. Accepted scales update $\log \bar{s}_j = (1-\eta) \log \bar{s}_{j-1} + \eta \log s_j$. Frames with too few inliers or excessive MAD are skipped and cannot vote.

C. Testing the Completed Map Across Views

The paired scan cannot label itself because it may contain the reflected returns being tested. We instead test fixed locations in the completed M_{raw} across historical views. This is also necessary because virtual returns move with viewpoint.

With voxel side length h, endpoint p_m^W has index $v_m = \lfloor p_m^W / h \rfloor$ and voxel centroid $\bar{p}_v^W$. We project the centroid into accepted frame j:

$$\begin{aligned} p_{jv}^C &= T_{CL} T_{LW}(t_j^D) \bar{p}_v^W, \\ (u_{jv}, z_{jv}^L) &= \big(\pi(K p_{jv}^C), [p_{jv}^C]_z\big). \end{aligned} \tag{7}$$

Here (u_{jv}, z_{jv}^L) are its pixel and camera-axis depth. A z-buffer keeps the nearest centroid per pixel. We reuse the border and depth-edge exclusions from scale calibration.

The calibrated prediction $\tilde{z}_{jv}^V = s_j D_j(u_{jv})$ estimates the nearest surface visible along the same camera ray. The signed range difference and its uncertainty margin are

$$r_{jv} = z_{jv}^L - \tilde{z}_{jv}^V, \qquad \delta_{jv} = r_{jv} - \max(\tau_a, \tau_r \tilde{z}_{jv}^V). \tag{8}$$

Positive r_{jv} places the voxel behind the visible surface. The required gap is the larger of absolute tolerance τ_a and relative tolerance $\tau_r \tilde{z}_{jv}^V$. Hence $\delta_{jv} > 0$ realizes the cue from Sec. I. A nearer surface should have blocked the endpoint. Predictions at or behind the voxel never support deletion.

D. Combining Evidence Across Viewpoints

Because prediction error, transmission, or occlusion can reproduce this ordering once, each frame casts at most one contradiction and one support vote:

$$\begin{aligned} g_j(v) &= \mathbf{1}[\delta_{jv} > 0], \\ a_j(v) &= \mathbf{1}[|r_{jv}| \le \tau_s]. \end{aligned} \tag{9}$$

Here $g_j(v)$ records contradiction and $a_j(v)$ agreement within tolerance τ_s. Per-frame voxel votes prevent point density from inflating evidence.

Let c_j be the camera center and $d_j(v)$ its direction to the voxel. To avoid recounting adjacent views, a new observation is accepted only if

$$\|c_j - c_{\text{last}}(v)\| \ge \tau_c \quad \vee \quad \angle(d_j(v), d_{\text{last}}(v)) \ge \tau_\theta. \tag{10}$$

Here "last" denotes the previous accepted view. A new vote therefore requires a change of at least τ_c in translation or τ_θ in viewing angle.

Let n_o count diverse visible views and n_c those with valid calibrated depth. Over valid checks, $n_g = \sum_j g_j(v)$ and $n_s = \sum_j a_j(v)$, with fractions $\rho_g = n_g/\max(1, n_c)$ and $\rho_s = n_s/\max(1, n_c)$.

Because the available calibrated frames F_{valid} vary by sequence, the minimum valid-check count adapts as

$$T_c(F_{\text{valid}}) = \min\left(T_c^{\max}, \max(2, \lceil \alpha F_{\text{valid}} \rceil)\right), \qquad (11)$$

where α is the required fraction and $T_c^{\max}$ is the cap. The geometric and support thresholds stay fixed.

A voxel is rejected only with enough observations, usable depths, and contradictions, but weak support:

$$\begin{aligned} C(v) = [n_o \geq T_o] \wedge [n_c \geq T_c(F_{\text{valid}})] \wedge [n_g \geq T_g] \\ \wedge [\rho_g \geq T_\rho] \wedge [\rho_s \leq T_s]. \end{aligned} \qquad (12)$$

Here T_o and T_g are minimum observation and contradiction counts, T_ρ is the minimum contradiction fraction, and T_s the maximum support fraction. Failure of any term retains the voxel. Thus viewpoint diversity blocks repeated single-view error, minimum counts block sparse evidence, and support protects repeated depth agreement. Candidates outside the camera view, occluded, or lacking calibrated depth are also retained. A physical object behind glass can still satisfy the ordering because the test concerns visible-surface consistency rather than material class.

Finally, the voxel decision is applied to the original endpoints:

$$M_{\text{clean}} = \{p_m^W \in M_{\text{raw}} : \neg C(\lfloor p_m^W / h \rfloor)\}. \qquad (13)$$

All native endpoints in a rejected voxel are removed. Retained endpoints are neither moved nor resampled.

IV. Experiments

A. Datasets

Our main dataset contains eight newly recorded reflective-elevator traversals (E01–E08) with synchronized RGB, Ouster LiDAR, and IMU. The traversals span seven measured cabin footprints. The complete set totals 265.2 s and 16.10 million frozen FAST-LIVO2 endpoints. For each sequence, one measured rectangle and height band are registered once in the raw-map frame and reused for every output. They are used only for evaluation, never cleaning, and provide a 2D reference rather than scanner-grade 3D ground truth.

3DRef [23] is used only for the reflection-mask control. A randomly initialized depthwise-separable encoder–decoder (16 base channels, eight epochs) is trained on its 2,850 official images and evaluated on the disjoint 949-image test split, without depth input. Its material labels are not used as map ground truth.

M2DGR's four lift sequences [1] test public lift integration. Hilti exp14 and exp18 [27] test a different LiDAR in non-elevator construction scenes. Neither dataset has compatible reflective-map labels, so we report execution, alignment coverage, and retention rather than ghost-removal accuracy.

B. Evaluation Protocol and Metrics

Let M_{elig} contain endpoints in a corridor extending 0.25 m inside and 1.20 m outside the measured rectangular boundary and in the declared height band, trimmed by 0.15 m at each end. At $\epsilon = 0.05$ m, the outside-boundary residual endpoint rate (RER) is the fraction of M_{elig} outside the cabin and farther than ϵ from its boundary. Wall precision is the fraction of M_{elig} within ϵ of a wall. Boundary coverage is the fraction of 2-cm perimeter samples supported by an endpoint within ϵ. Wall F-score is computed from precision and coverage for each sequence. Wall-distance P95 is the 95th percentile point-to-boundary distance over M_{elig}.

The exterior ghost footprint (EGF) measures spatial extent rather than point count. It is the area of occupied 10-cm cells in a cabin-aligned xy grid containing the exterior residual endpoints counted by RER. Occupied cells avoid filling gaps between disconnected artifacts. We interpret EGF together with coverage because deleting the entire map would also make it zero. These are 2D structural-shell metrics, not semantic ghost labels, 3D Chamfer distance, or trajectory ATE.

Full-map evaluation uses every output endpoint. For equal-count evaluation, each map is voxel-thinned without dropping below 50,000 points and then sampled to 50,000 points without replacement using seeds $\{11, 23, 37, 53, 71\}$. No map is upsampled. Seeds are averaged within each sequence, followed by an unweighted mean over eight traversals. F-score is computed before sequence averaging, so its reported mean need not equal the harmonic mean of reported mean precision and coverage. The two point-count settings are compared only within setting. The independent DA3 reconstruction is reported only at 50,000 points and is excluded from paired tests because it shares neither map, trajectory, nor endpoints with FAST-LIVO2.

For inference on the full-map FAST-LIVO2 pair, the seven physical footprints are the statistical units. The four one-sided exact Wilcoxon improvement tests are EGF, RER, wall precision, and wall F-score and are Bonferroni-adjusted. P95 and coverage are reported as descriptive geometric and preservation outcomes, respectively. Because the adaptive valid-check threshold and EGF were finalized after inspecting this campaign, the tests are exploratory rather than confirmatory. For matched-control inference, the predeclared primary outcome is the footprint-level F-score improvement of MirrorSentinel over random deletion, statistical outlier removal, radius outlier removal, and MirrorSentinel without temporal aggregation. We use four one-sided exact paired Wilcoxon tests over the seven footprints, with Holm correction. RER, P95, and coverage remain descriptive secondary outcomes. Public-sequence ATE is reported separately and is identical before and after cleaning because the trajectory is frozen.

C. Main Comparisons

The primary comparison enables or disables MirrorSentinel on the same frozen FAST-LIVO2 output [14]. The raw map, trajectory, registered scans, and endpoint coordinates are unchanged. The only difference is post-traversal cleaning.

We also replay FAST-LIVO2 with its mapping-time visual updates disabled. This "FAST-LIVO2 w/o visual updates" setting provides a LiDAR–IMU baseline, but is not an official FAST-LIO2 reproduction [10]. On these maps, we compare no cleaning, a vertical-plane filter adapted from Mapping with Reflection [4], and a fixed-vote MirrorSentinel preset. The plane method is an adaptation because our recordings lack its required dual-return input. These rows provide cross-system context and are not paired component ablations of the FAST-LIVO2 operating point.

The main comparison also includes an independent Depth Anything 3 (DA3) RGB-only baseline. Its DA3-BASE checkpoint reconstructs its own map and trajectory from 4-Hz, 504-pixel RGB without LiDAR, IMU, external poses, loop closure, floorplans, or ROIs. A memory-bounded adapter joins 120-frame windows with 60-frame overlap by confidence-weighted Huber-IRLS Sim(3). One global Umeyama Sim(3) [28] aligns its camera centers to the FAST-LIVO2 frame only for evaluation. This alignment cannot correct local shape or drift. Since MirrorSentinel instead uses DA3Metric-Large as a depth prior over frozen LiDAR geometry, DA3-BASE is a system-level baseline rather than a component ablation.

D. Ablations and Controls

To remove temporal aggregation without changing the deletion budget, each endpoint receives only its largest valid single-view margin,

$$s_i = \max_{j \in \mathcal{V}_i} \delta_{j v_i}, \tag{14}$$

where $\mathcal{V}_i$ is its set of valid historical checks and $s_i = -\infty$ when that set is empty. The no-temporal variant deletes the top K_s endpoints in sequence s, where K_s is exactly the number rejected by full MirrorSentinel. It does not use the aggregated counts or fractions in (12). The no-adaptive variant instead fixes $T_c = 6$ while retaining every other decision condition and its natural output size.

The matched controls operate on the same E01–E08 raw maps and use the same per-sequence K_s. Besides the no-temporal variant, they delete endpoints uniformly at random, by largest $k = 16$ mean-neighbor distance, or by smallest neighbor count within $r = 0.10$ m [29]. Random results are averaged over 50 seeds. We also test the support condition on the primary maps. The alternative depth priors and the reflection-mask, z-buffer, viewpoint-diversity, and voxel-size sensitivities use the visual-disabled FAST-LIVO2 maps. We do not pool these results with the primary operating point or its 50k outputs. Methods requiring full waveforms, 2D reflectance fields, or static multi-echo TLS remain Related Work comparisons [5]–[7], [25], [26].

The primary prior is `DA3Metric-Large` at 518×518 and 4 Hz. GemDepth and ZipDepth operate at 4 and 10 Hz, respectively. ZipDepth uses its official base checkpoint and inverse-depth output. Timing is measured on an RTX 4060 Laptop GPU with an i7-14700HX. Unless ablated, $\Delta t_{\max} = 0.05$ s, $h = 0.20$ m, $(\tau_a, \tau_r) = (0.40\,\mathrm{m}, 0.04)$, $(T_o, T_c^{\max}, T_g) = (2, 6, 1)$, $\alpha = 0.05$, $T_\rho = 0.50$, $(\tau_c, \tau_\theta) = (0.15\,\mathrm{m}, 5°)$, $\tau_s = 0.20$ m, and $T_s = 0.34$. Scale calibration uses a 3-pixel border, a 7×7 edge window with $\tau_e = 0.50$ m, $(\gamma_a, \gamma_r) = (0.30\,\mathrm{m}, 0.05)$, $\lambda = 0.20$, $\kappa = 3.5$, $\eta = 0.20$, scale range $[10^{-3}, 10^3]$, and $|\log(s/\bar{s})| \leq 0.50$. A frame requires at least 30 inliers and calibrated residual MAD at most 0.50 m. No parameter depends on a floorplan or target point count. Run manifests record commands, effective parameters, source hashes, synchronization, and per-frame calibration diagnostics. A release-code replay from the frozen maps, trajectories, priors, and configuration reproduces all eight natural clean maps, rejected maps, vote-statistic files, and accepted-frame lists byte for byte.

V. Results

A. Main Results

On the same frozen FAST-LIVO2 maps, RER, P95, and F-score improve on all eight traversals. Coverage ties on seven and decreases by 0.0026 on one. Across all points, mean RER decreases from 0.3968 to 0.3465 (12.7%) and P95 from 0.8158 to 0.6230 m (23.6%). Precision and F-score increase, while coverage changes from 0.8356 to 0.8353. Mean EGF decreases from 4.1900 to 2.4275 m^2 (42.1%). At 50,000 points, it decreases from 3.2748 to 2.2435 m^2 (31.5%). After grouping E02 and E03, the median paired improvements are 0.0254 in RER, 0.135 m in P95, 0.0211 in precision, and 0.0211 in F-score. All seven footprint effects improve for EGF, RER, precision, F-score, and P95. For the four tested outcomes (EGF, RER, precision, and F-score), the exploratory one-sided exact Wilcoxon p-value is 0.0078 and the Bonferroni-adjusted value is 0.0313. P95 is descriptive because it was not included in that four-test family.

Table I reports the full-map means and comparisons at 50,000 points.

The 50k rows in Table I compare methods at the same point count. Enabling MirrorSentinel on FAST-LIVO2 decreases RER from 0.6330 to 0.4943 and P95 from 1.1203 to 0.8452 m, while precision, coverage, and F-score increase. On FAST-LIVO2 w/o visual updates, the plane filter reaches lower RER than the fixed-vote setting (0.6386 versus 0.6826) but reduces coverage from 0.8137 to 0.5185. The fixed-vote MirrorSentinel setting reaches the highest F-score and coverage of the three visual-disabled rows. The independent RGB-only DA3-BASE baseline obtains RER/F-score/coverage of 0.8043/0.1105/0.4501 after its evaluation-only alignment. Because it also estimates poses and joins temporal windows, this is system-level context rather than a component test of MirrorSentinel.

On the RTX 4060, DA3Metric-Large, GemDepth, and ZipDepth model calls run at 12.5, 9.8, and 135.3 FPS, respectively. These are model-call throughputs, not complete online-SLAM rates. A serial CPU replay cleans all eight sequences in 2517.9 s (314.7 s per sequence), excluding mapping and prior generation.

Figure 4 shows the spatial effect on all eight traversals.

B. Matched Controls

Figure 3 and Table II compare six endpoint decisions on the same frozen maps under identical per-sequence rejection

TABLE I

E01–E08 SEQUENCE MEANS. FULL-MAP ROWS COMPARE PAIRED FROZEN MAPS. THE 50K ROWS GIVE EQUAL-COUNT CROSS-SYSTEM CONTEXT. †EVALUATION-ONLY SIM(3).

Method	RER↓	P95 [m] ↓	Precision ↑	Coverage ↑	F-score ↑
Full maps					
FAST-LIVO2 w/o MirrorSentinel	.3968	.8158	.4310	**.8356**	.5448
FAST-LIVO2 w/ MirrorSentinel	**.3465**	**.6230**	**.4607**	.8353	**.5757**
50k points					
FAST-LIVO2 w/o visual updates	.7177	1.1053	.1778	.8137	.2848
w/ plane filter	.6386	1.0988	.2073	.5185	.2887
w/ MirrorSentinel fixed vote	.6826	1.0943	.2083	**.8722**	.3301
DA3-BASE (RGB only)†	.8043	1.0699	.0651	.4501	.1105
FAST-LIVO2 w/o MirrorSentinel	.6330	1.1203	.2471	.7357	.3537
FAST-LIVO2 w/ MirrorSentinel	**.4943**	**.8452**	**.3368**	.8205	**.4619**

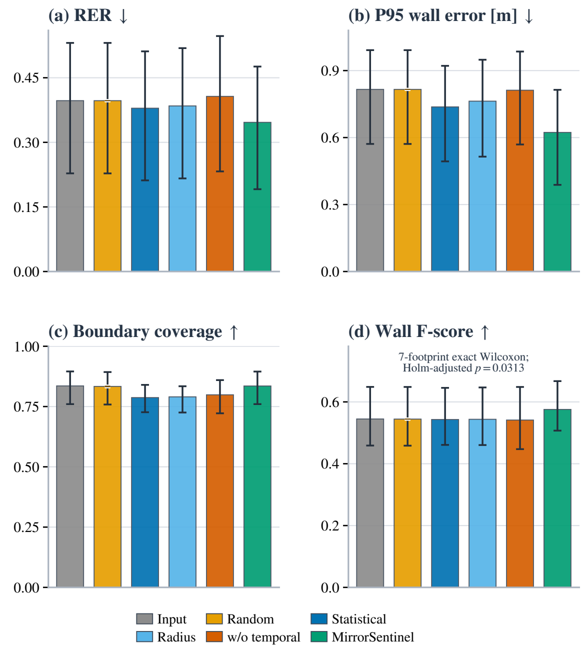


Fig. 3. Exact reject-count controls on the same frozen FAST-LIVO2 maps. Bars show unweighted E01–E08 sequence means. Outer whiskers are descriptive 95% cluster-bootstrap intervals over seven footprints (20,000 replicates). Random inner caps show the 5th–95th percentiles across 50 aggregate seed means. Only F-score is used for inference. The four predeclared footprint-level tests give Holm-adjusted $p = 0.0313$.

budgets. MirrorSentinel obtains the lowest mean RER and P95 and the highest mean F-score while preserving input-map boundary coverage. Its footprint-level F-score is higher on 7/7 footprints than random, statistical, and no-temporal controls and on 6/7 than radius removal. All four Holm-adjusted tests give $p = 0.0313$.

Statistical and radius filtering reduce EGF slightly further (2.243/2.233 versus 2.428 m^2), but lose substantially more boundary coverage. The matched results therefore favor multi-view contradiction and support evidence over indiscriminate sparsification or a single residual ranking.

TABLE II

NATURAL-OUTPUT MEANS UNDER PER-SEQUENCE MATCHED REJECTION BUDGETS. RANDOM IS AVERAGED OVER 50 SEEDS.

Method	RER↓	P95[m]↓	Cov.↑	F↑
Random	.397	.816	.833	.544
Statistical	.379	.737	.787	.543
Radius	.385	.763	.790	.544
w/o temporal	.407	.812	.799	.541
MirrorSentinel	**.346**	**.623**	**.835**	**.576**

TABLE III

TEMPORAL-EVIDENCE ABLATION OVER E01–E08. REJECT IS THE MEAN FRACTION OF NATIVE ENDPOINTS REMOVED.

Method	Reject	RER↓	P95[m]↓	Cov.↑	F↑
Input map	.000	.397	.816	**.836**	.545
MirrorSentinel w/o temporal	.274	.407	.812	.799	.541
MirrorSentinel w/o adaptive	.263	.351	.629	.835	.573
MirrorSentinel (full)	.274	**.346**	**.623**	.835	**.576**

C. Ablation of Temporal Evidence Aggregation

Table III separates the contribution of temporal aggregation from deletion count and adaptive thresholding. The input, no-adaptive, and full rows use their natural outputs. The no-temporal row uses the same per-sequence rejection count as full MirrorSentinel.

At the same 0.274 rejection fraction, the no-temporal variant lowers coverage from 0.835 to 0.799 and F-score from 0.576 to 0.541. Its F-score is also below the unfiltered input. A single large visual–LiDAR discrepancy is therefore not a reliable deletion rule. Fixing $T_c = 6$ recovers most of the gain, while the adaptive threshold supplies a smaller improvement of 0.0047 RER and 0.0024 F-score. Removing the support condition gives RER/coverage/F-score of 0.344/0.828/0.575, indicating a small, metric-dependent preservation effect rather than a primary accuracy gain.

On the visual-disabled FAST-LIVO2 maps, the depth-prior comparison holds the raw map and decision rule fixed. DA3Metric-Large has the highest accepted-frame fraction on six of eight traversals and obtains RER/F-score 0.608/0.425, versus 0.621/0.414 for GemDepth and 0.620/0.415 for ZipDepth.

The 3DRef RGB head reaches IoU/F1 of 0.628/0.771, but

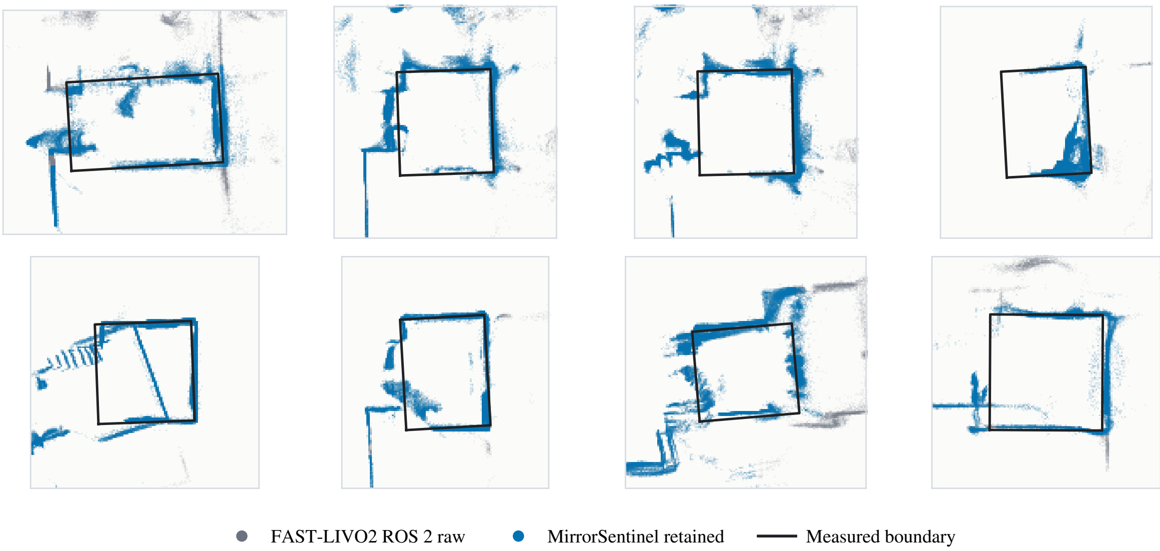


Fig. 4. Natural-output maps for E01–E08. Gray shows each raw FAST-LIVO2 map, and blue overlays retained endpoints. Exposed gray therefore marks removals. Black boundaries are used only for display and evaluation. Cleaning uses neither floorplans nor ROIs, and all retained coordinates are original FAST-LIVO2 endpoints.

requiring its mask changes GemDepth RER/F-score from 0.621/0.414 to 0.627/0.409. On the same maps, removing the z-buffer or viewpoint diversity gives 0.622/0.414 or 0.611/0.422, respectively. For 10–30 cm voxels, RER remains 0.620–0.623 and F-score 0.413–0.415.

D. Public-Sequence Results

All six public streams complete. M2DGR accepts 45.6–78.4% of aligned frames, retains 69.1–79.2% of endpoints, and has shared raw/clean tracker ATE of 1.042–3.448 m. We use translation-only association because the released lift quaternions are zero. Hilti exp14 calibrates 14/95 frames, retains 91.3%, and has shared ATE 0.159 m. Exp18 calibrates 11/117 frames and retains every point. Its 4.222-m ATE violates the trajectory precondition, making this run a full abstention. Without reflective-map labels, these runs demonstrate integration and retention, not cleaning accuracy.

VI. Discussion and Limitations

The same-map experiment isolates post-mapping endpoint selection from trajectory estimation, and the 50k comparison shows that the gain persists after output-density normalization. The per-sequence controls show which points matter. At the same rejection budget, random removal largely preserves the raw-map error, generic outlier filters trade boundary support for a smaller footprint, and the largest single-view residual lowers F-score below the unfiltered input. Temporal aggregation explains most of the gain. Repeated, viewpoint-conditioned evidence recovers the cabin boundary while removing exterior residuals. Fixing the evidence threshold at six retains most of this gain, whereas adaptation provides a smaller improvement for short valid-evidence streams. The support condition is similarly conservative rather than a primary accuracy source. Freezing the trajectory prevents map edits from changing registration, but it does not make the cleaner immune to a systematically wrong depth prior.

The estimated property is consistency with the nearest visible surface rather than material or reflection class. This distinction defines both the utility and the main failure mode. A physical object behind transparent glass can be inconsistent with the nearer visible surface and may therefore be undesirable in a structural-shell export but essential in an obstacle map. Likewise, a ghost that remains outside the camera frustum, is repeatedly occluded, or is followed by the depth model will accumulate insufficient evidence and be retained. MirrorSentinel should consequently be viewed as an offline structural-map cleaner for a usable completed trajectory, not online pose recovery, semantic reflection classification, or a safety-critical navigation filter.

The evaluation is exploratory. Eight recorded sequences cover seven cabin footprints from one campaign. The adaptive count and footprint metric were finalized after inspecting that campaign, and uncertainty in the manually registered 2D floorplans is not quantified. The public M2DGR and Hilti runs demonstrate execution and abstention only because compatible reflection labels are absent. Stronger evidence would require an unseen elevator with frozen parameters, nonreflective false-deletion scenes, floorplan-perturbation analysis, and semantic or scanner-grade 3D references.

VII. Conclusion

MirrorSentinel uses calibrated visual depth to repeatedly check whether a mapped voxel lies behind the nearest visible surface. Across eight elevator sequences, it reduces ghost footprint by 42.1% while increasing wall F-score with only -0.0003 mean coverage change. The reduction persists under an exact 50,000-point density normalization. Because poses and retained coordinates remain unchanged, the result is conservative offline map cleaning rather than odometry or visual reconstruction. Under identical per-sequence rejection budgets, it also outperforms random, statistical, radius, and single-view residual controls in footprint-level F-score (Holm-adjusted $p = 0.0313$). The matched single-view ablation further identifies temporal aggregation, rather than visual depth alone, as the main decision mechanism. A prospectively frozen unseen elevator is the next test of transfer.

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